

A permutation-fair collection with one polynomial-size die

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Status. Complete proof with independent mathematical reviews completed by three collaborating agents. These reviews are not machine-checked formal verification. The theorem disproves the manuscript's conjecture that every die in a permutation-fair collection must have exponentially many faces. The other dice in the construction can be arbitrarily large; this does not improve the optimum total size.

Theorem 1. *There is an absolute constant C such that, for every $n \geq 2$, there is a permutation-fair collection of n finite uniform dice, with pairwise disjoint face labels, in which one distinguished die has at most Cn^2 faces. The other $n - 1$ dice can all be chosen with the same number of faces.*

The argument first builds positive rational piecewise-constant densities for the other dice and a finite uniform distinguished die. It then replaces each constant-density interval by a finite permutation-fair block.

Write $\mathcal{P}_d$ for the real polynomials of degree at most d .

Lemma 2 (Regular real starting rule). *For every $m \geq 1$ there is a real equal-weight uniform-interval quadrature of degree m with*

$$m^2 \leq K = O(m^2), \quad 0 < u_1 < \cdots < u_K < 1.$$

Proof. The classical Chebyshev-type quadrature bound gives an equal-weight degree- $(2m + 2)$ rule with $K_0 = O(m^2)$ nodes in $[0, 1]$. For example, use the uniform-weight case of Gilboa–Peled, Theorem 1.1: <https://arxiv.org/abs/1507.01505>. It has at least $m + 2$ distinct support points: otherwise the square of its support-vanishing polynomial would have degree at most $2m + 2$, zero quadrature sum, and positive integral. At least m support points are interior. Repeat the rule $\lceil m^2/K_0 \rceil$ times so its cardinality K is at least m^2 and still $O(m^2)$.

Select m coordinates at distinct interior support points as pivots. The Jacobian of the first m moment sums with respect to them is an invertible scaled Vandermonde matrix. The implicit function theorem permits arbitrary small perturbations of the remaining coordinates, compensating with the pivots to preserve these moment sums. The pivots remain interior and mutually distinct. Move every free endpoint slightly inward.

Free-free collisions are proper affine hyperplanes. If a free coordinate and a pivot coincide initially, varying just that free coordinate changes its matching pivot with derivative -1 , since their Jacobian columns coincide; all other pivot derivatives are zero. Their difference therefore has derivative -2 , so that collision is a proper analytic zero set. Initially absent collisions stay absent in a sufficiently small neighborhood. A finite union of proper analytic zero sets cannot cover the open set of permitted inward perturbations. Choose a collision-free perturbation, then sort the nodes. $\square$

Lemma 3 (Rational separated moment bumps). *For the nodes in Lemma 2, one can choose pairwise disjoint open intervals $I_q \subset (0, 1)$ around u_q , smaller neighborhoods $V_q \subset I_q$ around u_q , and bounded rational piecewise-constant functions h_q , with rational breakpoints and support in I_q , such that for every $t \in V_q$ and every $g \in \mathcal{P}_{m-1}$,*

$$\int_0^t h_q(x)g(x) dx = g(0), \quad \int_t^1 h_q(x)g(x) dx = -g(0).$$

In particular $\int_0^1 h_q(x)g(x) dx = 0$.

Proof. Choose disjoint rational-endpoint neighborhoods I_q . Inside each I_q , choose m disjoint rational closed subintervals strictly to the left of u_q , and m strictly to its right, all of positive length. Their union misses a neighborhood V_q of u_q .

For any m ordered disjoint intervals $J_1 < \dots < J_m$, the rational matrix

$$M_{\ell a} = \int_{J_a} x^\ell dx \quad (0 \leq \ell \leq m-1, 1 \leq a \leq m)$$

is invertible. Indeed, multilinearity of the determinant gives

$$\det M = \int_{J_1 \times \dots \times J_m} \prod_{a < b} (x_b - x_a) dx_1 \dots dx_m > 0.$$

Solve $M\gamma = (1, 0, \dots, 0)^T$ for the left intervals and $M\gamma = (-1, 0, \dots, 0)^T$ for the right intervals. Use these rational coefficients as the constant values of h_q on those intervals, and zero elsewhere. The claimed identities follow by linearity. No nonnegativity is claimed for h_q itself. $\square$

Lemma 4 (Order response of disjoint moment bumps). *Choose $t_q \in V_q$. Let $Q_1, \dots, Q_m$ be pairwise disjoint subsets of $[K]$. Suppose*

$$f_j(x) = 1 + \sum_{q \in Q_j} \delta_{j,q} h_q(x), \quad j \in [m],$$

are nonnegative densities. Let X_j have density f_j , independently, and let X_0 be independent and uniform on $t_1, \dots, t_K$. Define functionals on $\mathcal{P}_{m-1}$ by

$$B_j(a) = \frac{1}{K} \sum_{q \in Q_j} \delta_{j,q} a(t_q).$$

If all B_j agree with one functional B , then every full ordering placing X_0 after exactly k of the old variables has probability

$$\frac{1}{K} \sum_{q=1}^K p_k(t_q) + B(p'_k), \quad p_k(t) = \frac{t^k(1-t)^{m-k}}{k!(m-k)!}. \quad (1)$$

Proof. Fix a full order and condition on $X_0 = t_\ell$. Expand the product of the old densities. The term choosing no bumps has probability $p_k(t_\ell)$.

Consider a term choosing bumps for a nonempty set of old variables, and expand each chosen density's sum over its indices. All chosen bumps have distinct indices because the Q_j are disjoint. Their supporting intervals are therefore disjoint and ordered. If their order is inconsistent with the fixed full order, the term is zero. Otherwise their mutual ordering is automatic throughout those intervals. Integrate out the old variables whose density contribution was the constant 1. If there are b chosen bumps, the resulting kernel is a product of powers of the gaps between the chosen coordinates, 0, 1, and t_ℓ , divided by factorials. It is a polynomial of total degree $m - b$ in the chosen coordinates and t_ℓ on each of the relevant regions.

At most one supporting interval contains t_ℓ , namely I_ℓ . If $b \geq 2$, at least one chosen bump has its entire support on one side of t_ℓ . For fixed remaining coordinates, its integration runs over its entire support against a polynomial of degree at most $m - b \leq m - 1$. The full moment identities of Lemma 3 make this integral zero. The same argument eliminates a one-bump term whose index is not ℓ . Thus only the one-bump terms cut by the distinguished face remain.

We calculate them. If the perturbed old variable is before X_0 , let a old variables precede it and b lie between it and X_0 , so $a + b = k - 1$. Its kernel is

$$g_{a,b}^-(x, t) = \frac{x^a(t-x)^b(1-t)^{m-k}}{a!b!(m-k)!}.$$

Its integration against the left part of h_q is $g_{a,b}^-(0, t)$. If the perturbed variable is after X_0 , let a old variables lie between them and b follow it, so $a + b = m - k - 1$. Its kernel is

$$g_{a,b}^+(x, t) = \frac{t^k(x-t)^a(1-x)^b}{k!a!b!},$$

whose integral against the right part of h_q is $-g_{a,b}^+(0, t)$. These evaluated kernels have degree at most $m - 1$ in t .

Average over X_0 and sum over the old variables. Since all $B_j = B$, the resulting correction is B applied to the sum of these kernel polynomials. The binomial theorem gives

$$\begin{aligned} \sum_{a+b=k-1} g_{a,b}^-(x, t) &= \frac{t^{k-1}(1-t)^{m-k}}{(k-1)!(m-k)!}, \\ \sum_{a+b=m-k-1} g_{a,b}^+(x, t) &= \frac{t^k(1-t)^{m-k-1}}{k!(m-k-1)!}. \end{aligned}$$

Use the first expression only when $k \geq 1$ and the second only when $k \leq m - 1$. Their difference is $p'_k(t)$, including the endpoint cases. This proves (1). $\square$

Proof of Theorem 1: continuous intermediate model. Put $m = n - 1$ and choose the regular real rule and bumps above. Since $K \geq m^2$, choose disjoint sets $Q_1, \dots, Q_m$, each of size m . Choose rational $t_q \in V_q$ arbitrarily close to u_q . Their ordering remains strict. For $1 \leq s \leq m$, put

$$\mu_s(t) = \frac{1}{K} \sum_{q=1}^K t_q^s, \quad a_s(t) = \frac{1/(s+1) - \mu_s(t)}{s}.$$

For each j , solve the rational Vandermonde system

$$\frac{1}{K} \sum_{q \in Q_j} \delta_{j,q} t_q^{s-1} = a_s(t) \quad (1 \leq s \leq m).$$

The matrix is invertible because the t_q are distinct. The solution depends continuously on t near u and tends to zero as $t \rightarrow u$, because $a_s(u) = 0$. All h_q are fixed bounded functions. Taking the rational approximation sufficiently close therefore guarantees $f_j(x) \geq 1/2$ everywhere, where

$$f_j(x) = 1 + \sum_{q \in Q_j} \delta_{j,q} h_q(x).$$

Their integrals are 1 by the full zeroth-moment identity for the bumps. They are thus positive rational piecewise-constant probability densities.

The functionals B_j agree on the monomial basis of $\mathcal{P}_{m-1}$, and hence agree with one functional B . For every $p \in \mathcal{P}_m$,

$$B(p') = \int_0^1 p(t) dt - \frac{1}{K} \sum_{q=1}^K p(t_q),$$

as follows by checking $p(t) = t^s$, $1 \leq s \leq m$, and constants. Apply Lemma 4 to p_k . Every full order has probability

$$\int_0^1 p_k(t) dt = \frac{1}{(m+1)!}.$$

This proves permutation fairness for the continuous old variables and the K -face distinguished variable. $\square$

Finite realization, completing the theorem. Partition $[0, 1]$ at every rational breakpoint of every f_j and at all t_q . Ignore singleton boundary points, which have probability zero for the old variables. On each remaining interval J_g , each f_j is a positive rational constant. The masses

$$\alpha_{j,g} = \int_{J_g} f_j(x) dx$$

are positive rationals, and $\sum_g \alpha_{j,g} = 1$ for each j . Choose a common positive integer D clearing all their denominators.

Take any finite equal-sided permutation-fair collection of m dice, with F faces each, whose existence is already proved in the manuscript. In interval J_g , put a copy of its fair word in which letter j is blown up by the positive integer $D\alpha_{j,g}$. This block is permutation-fair, and contains $FD\alpha_{j,g}$ faces of old die j . Concatenate the blocks in interval order, with one distinguished face inserted at each boundary t_q . Every old die has exactly FD faces; the distinguished die has exactly K .

For each old die, its interval probabilities equal $\alpha_{j,g}$, exactly as in the continuous model. Conditional on the selected intervals, old variables in distinct intervals have a determined relative order. Those in a common interval are uniformly ordered in the continuous model, because their conditional distributions are all uniform on that interval. They are also uniformly ordered in the discrete model, by permutation fairness of that block and its restrictions. Different interval groups involve disjoint sets of independent dice, so their internal orders combine independently. Thus the entire order law, including the distinguished faces at the interval boundaries, is unchanged. Assign successive distinct integer labels to the concatenated word to obtain the required finite dice. $\square$

What is and is not proved. The distinguished die has $O(n^2)$ faces, while the sizes of the other dice are not bounded quantitatively here. The construction uses arbitrarily accurate rational approximation and correspondingly large denominator clearing only for those other dice. It does not conflict with the manuscript's result that almost all dice must be exponentially large, nor with the new exponential obstruction for rational scalar equal-weight interval designs. The local CDF vectors are deliberately different for different old dice.